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Score: _____

Directions: Put away phones, calculators, smart watches, etc.

Answer in the space provided.

Show work.

1. For this problem, $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 5 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -4 & 3 \end{bmatrix}$, $C = [-2 \ 4]$, and $D = [-2 \ 0]$.

Perform the indicated operations or state that they are not possible.

$$(a) \quad BA = \begin{bmatrix} 2 & 1 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 2 & 3 & -1 \\ 1 & 5 & 5 \end{bmatrix} = \boxed{\begin{bmatrix} 5 & 11 & 3 \\ -5 & 3 & 19 \end{bmatrix}}$$

$$(b) \quad CBC^T = [-2 \ 4] \begin{bmatrix} 2 & 1 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix} = [-20 \ 10] \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \boxed{[80]}$$

$$(c) \quad B^{-1} = \frac{1}{2 \cdot 3 - (-4) \cdot 1} \begin{bmatrix} 3 & -1 \\ 4 & 2 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 3 & -1 \\ 4 & 2 \end{bmatrix} = \boxed{\begin{bmatrix} \frac{3}{10} & -\frac{1}{10} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}}$$

- (d) Find X if $C^T D - 2X^T = I_2$.

$$\begin{bmatrix} -2 \\ 4 \end{bmatrix} \begin{bmatrix} -2 & 0 \end{bmatrix} - 2X^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 \\ -8 & 0 \end{bmatrix} - 2X^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 0 \\ -8 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 2X^T$$

$$2X^T = \begin{bmatrix} 3 & 0 \\ -8 & -1 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 3/2 & 0 \\ -4 & -1/2 \end{bmatrix}$$

$$X = \boxed{\begin{bmatrix} 3/2 & -4 \\ 0 & -1/2 \end{bmatrix}}$$

2. Suppose A , B and C are 3×3 matrices, and $\det(A) = 5$, and $\det(B) = 3$.

$$(a) \quad \det(A^2 B) = \det(A^2) \det(B) = \det(A) \det(A) \det(B) = 5 \cdot 5 \cdot 3 = \boxed{75}$$

$$(b) \quad \det(2B^{-1}) = 2^3 \det(B^{-1}) = \frac{8}{\det(B)} = \boxed{\frac{8}{3}}$$

- (c) Suppose $\det(ABC) = 10$. Find $\det(C)$.

$$\det(A) \det(B) \det(C) = 10$$

$$5 \cdot 3 \det(C) = 10 \rightarrow \det(C) = \frac{10}{15} = \boxed{\frac{2}{3}}$$

3. Solve the system
$$\begin{cases} -3w + 6x - 3y - 9z = 12 \\ w - 2x + y + 3z = -4 \\ -\frac{1}{3}w + \frac{2}{3}x + \frac{1}{3}y - z = \frac{4}{3} \end{cases}$$

$$\begin{bmatrix} -3 & 6 & -3 & -9 & 12 \\ 1 & -2 & 1 & 3 & -4 \\ -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} & -1 & \frac{4}{3} \end{bmatrix}$$

$$\begin{aligned} -\frac{1}{3} R_1 \rightarrow R_1 \\ 3 R_3 \rightarrow R_3 \end{aligned} \begin{bmatrix} 1 & -2 & 1 & 3 & -4 \\ 1 & -2 & 1 & 3 & -4 \\ -1 & 2 & 1 & -3 & 4 \end{bmatrix}$$

$$\begin{aligned} -R_1 + R_2 \rightarrow R_2 \\ R_1 + R_3 \rightarrow R_3 \end{aligned} \begin{bmatrix} 1 & -2 & 1 & 3 & -4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \frac{1}{2} R_3 \rightarrow R_3 \\ R_2 \leftrightarrow R_3 \end{aligned} \begin{bmatrix} 1 & -2 & 1 & 3 & -4 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$-R_2 + R_1 \rightarrow R_1 \begin{bmatrix} 1 & -2 & 0 & 3 & -4 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{reduced row echelon})$$

$$\begin{cases} w - 2x + 3z = -4 \\ y = 0 \end{cases}$$

Solution:

$$\begin{aligned} w &= -4 + 2s - 3t \\ x &= s \\ y &= 0 \\ z &= t \end{aligned} \quad (s, t \in \mathbb{R})$$

4. Find x such that the matrix $\begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}$ is equal to its own inverse.

Need $\begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}^{-1} = \begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}$

$$\frac{1}{(3)(-3) - (-2)x} \begin{bmatrix} -3 & -x \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}$$

$$\frac{1}{2x-9} \begin{bmatrix} -3 & -x \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}$$

$$\frac{-1}{2x-9} \begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}$$

$$\therefore \frac{-1}{2x-9} = 1$$

$$-1 = 2x - 9$$

$$8 = 2x$$

$$\boxed{x = 4}$$

5. Find the determinant of the matrix $A = \begin{bmatrix} 3 & 5 & 2 & 4 \\ 1 & 2 & 5 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.

Expand along row 4:

$$|A| = - \begin{vmatrix} 3 & 5 & 2 \\ 1 & 2 & 5 \\ 0 & 1 & 2 \end{vmatrix}$$

Now expand this along row 3

$$|A| = - \left(0 \begin{vmatrix} 5 & 2 \\ 2 & 5 \end{vmatrix} - \begin{vmatrix} 3 & 2 \\ 1 & 5 \end{vmatrix} + 2 \begin{vmatrix} 3 & 5 \\ 1 & 2 \end{vmatrix} \right)$$

$$= - \left(- (15 - 2) + 2(6 - 5) \right)$$

$$= - (-13 + 2)$$

$$= \boxed{11}$$